

Machine Learning Applications: Complete Classroom Notes

Course: Machine Learning (Day 8)

Teacher: CampusX

Focus: Real-Life B2B and B2C Applications

1. Introduction: Why Study ML Applications?

The teacher emphasizes that before diving into the "heavy" mathematics and coding, it is vital to understand the **impact** of Machine Learning (ML).

Key Concept: ML is Everywhere

We already use ML daily, often without realizing it.

- **Facebook:** Friend recommendations.
- **YouTube/Amazon:** Product and video recommendations.
- **Current Reality:** ML is not "the future"; it is already integrated into our lives through hundreds of products.

2. B2C vs. B2B Applications

The teacher introduces a critical distinction in how ML is used:

1. **B2C (Business to Customer):** Standard applications where a business interacts with a normal user (e.g., WhatsApp, YouTube).
2. **B2B (Business to Business):** This is where one business helps another business run more efficiently using ML.
 - *Teacher's Tip:* B2B applications often generate significantly more revenue than B2C.

3. Sector 1: Retail Sector

Retail is one of the biggest users of ML. Without it, modern e-commerce would fail.

A. Inventory Management (Example: Amazon)

- **Scenario:** During sales like the "Great Indian Festival," Amazon sells over 6 crore (60 million) products.
- **The Problem:** Amazon cannot increase the stock of every single product. That would waste billions of dollars.
- **The ML Solution:** Data Scientists analyze years of past sales data and the current sale's trends to predict **which specific products will sell**.

- **Result:** They only stock up on items predicted to have high demand, saving money and storage space.

B. Customer Profiling & Targeted Marketing (Example: Big Bazaar/Spencers)

- **The Process:** When you shop, stores ask for your **phone number**.
- **Why?** It's not just for SMS alerts. They are tracking your **Buying Behavior**.
- **The Profiling:**
 - If you buy oats, milk, and gym supplements → Profile: **Health Conscious**.
 - If you buy diapers and baby powder → Profile: **New Parent**.
- **Monetization:** This profile data is sold to advertisers. A gym owner will pay more to get the phone numbers of 100 "Health Conscious" people than to send 1,000 random SMS messages. This is called **Targeted Marketing**.

C. Market Basket Analysis (Positioning)

- **Logic:** ML algorithms (like Association Rule Mining) calculate the correlation between products.
- **Example:** If people who buy Diapers also tend to buy Beer, the store will place them near each other to increase sales.

4. Sector 2: Banking & Finance

A. Loan Approval & Risk Assessment

- **The Process:** When you apply for a loan, your profile is compared against a database of "Past Defaulters" (people who didn't pay back).
- **Logic:** If your profile matches a defaulter's profile (e.g., income level, job stability, age), the ML model assigns a "Probability of Default."
- **Result:** If the probability is high (e.g., 83%), the bank's system rejects your application automatically.

B. Other Uses:

- Deciding where to open new branches.
- Creating city-specific promotional plans.
- **Algorithmic Trading:** Using ML to buy/sell stocks automatically.

5. Sector 3: Transportation & Logistics

A. Surge Pricing (Example: Uber/Ola)

- **Concept:** "Demand-Supply Forecasting."
- **How it works (Driver's Side):** The Driver's app shows "Red Zones." If a driver enters a Red Zone, Uber offers them an incentive (e.g., 2x or 3x the normal pay).
- **How it works (User's Side):** Since Uber is paying the driver more to come to your

high-demand area, they charge you a "Surge Price."

- **The Logic:** This maintains the supply of cabs in areas where demand is high.

B. Logistics Optimization

- Companies like **Delhivery** use ML to find the most "optimized route" for a delivery boy carrying 30 packages to ensure the fastest delivery with the least fuel.

6. Sector 4: Manufacturing

A. Predictive Maintenance (Example: Tesla Factories)

- **Scenario:** Tesla factories use robotic arms. If one arm breaks, the whole production line stops, causing millions in losses.
- **The Solution:** They use **IoT Sensors** on the robots.
- **Logic:** A robot doesn't usually fail suddenly. It fails gradually. Sensors monitor:
 - Temperature
 - RPM (Revolutions Per Minute)
 - Pressure
- **Warning Sign:** If the normal RPM of 300 drops slowly to 295... 290... 285, the ML model predicts a failure is coming.
- **Action:** Engineers fix it *before* it breaks. This is called **Predictive Maintenance**.

7. Sector 5: Consumer Internet (Twitter Sentiment Analysis)

The teacher shares a fascinating case study on how Twitter became profitable.

Concept: Sentiment Analysis

Definition: Using ML to read text and determine if the emotion/opinion is **Positive** or **Negative**.

The Sentiment Analysis Tool (Logic Explanation)

The teacher shows a tool he built. Here is the logic flow:

1. **Input:** Search for a movie (e.g., *Dunkirk*).
2. **Scraping:** The code goes to IMDB and fetches all user reviews.
3. **Processing:** The ML model reads each review.
4. **Classification:**
 - "Brilliant cinematography" → **Positive (Green)**
 - "I was disappointed/cheated" → **Negative (Red)**
5. **Output:** A final rating based on the ratio of Good to Bad reviews.

Twitter's Smart Business Plan (Case Study)

- **The Problem:** Twitter was not making enough money from ads.

- **The ML Solution:** They started selling **Real-Time Sentiment Data**.
- **Example (Elections):** * Twitter tracks hashtags like #Elections2024.
 - They analyze 50,000 tweets to see who people *actually* support.
 - This "Opinion Poll" is extremely accurate because people express their real feelings on social media.
- **The "Secret" Buyers:** They don't just sell to news channels; they sell to **Stock Brokers**.
- **Why Stock Brokers?** If brokers know a "pro-business" government is likely to win (based on Twitter sentiment), they buy stocks of companies that would benefit (e.g., Reliance). When the government actually wins, stock prices soar, and they make millions in profit. Twitter gets a "cut" for providing the intelligence.

8. Summary & Motivation

- **Why Learn This?** To become a professional who can build tools that change how companies operate.
- **Teacher's Encouragement:** When math or coding gets hard in the coming days, remember these applications. ML gives you a "Power Tool" to solve real-world problems and even start your own company.

Key Rule to Remember: > "If you are not paying for the product, you ARE the product." (Your data is the real value).